

# Towards an Immune System for Large Language Models: The Five-Anchor Audit Framework and Four-Tier Detection Chain

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## Abstract

As large language models (LLMs) become embedded in critical decision-making pipelines, the threat of coordinated data poisoning has escalated from a theoretical concern to an active attack surface. Current defenses—RLHF, content filtering, adversarial training—are passive, reactive, and fundamentally un-auditable. We propose a paradigm shift: an *immune system* for AI that detects, contains, and learns from poisoned data through a multi-layered framework consisting of (1) a Four-Tier Detection Chain that escalates from publisher credibility to cross-cluster gang identification, (2) a Five-Anchor Audit system that prevents single-point verification failures, and (3) a Three-Line Defense architecture that closes the loop from metadata to physical reproducibility. Unlike prior work that treats data poisoning as input sanitization, our framework treats it as an immune challenge—the system learns from each exposure, building calibration memory over time. We open-source the detection methodology and audit standards to enable community validation and iterative hardening.

## 1 Introduction

The volume of AI-generated content is growing exponentially. So is the volume of AI-generated *poison*. Coordinated data poisoning attacks are no longer carried out by individual bad actors inserting single artifacts; they are now industrialized operations producing self-referential networks of fabricated papers, datasets, and citation chains that appear internally consistent but are systematically detached from empirical reality [1].

Current approaches to AI safety fall into three categories: reinforcement learning from human feedback (RLHF), content filtering pipelines, and adversarial training. All three share a structural weakness: they operate on the *output* side of the model, after the training data has already been ingested. They are passive filters, not active immune systems. They cannot answer the question: *how do we know this training data point was real?*

We argue that what is needed is not better filtering, but a fundamentally different architecture—an **immune system** that:

1. Detects poisoned data *before* it enters the training pipeline
2. Provides auditable evidence chains for every detection decision
3. Learns from each exposure to build calibration memory
4. Allows false positives to be challenged and corrected
5. Closes the loop through physical reproducibility verification

This paper presents the design, methodology, and initial calibration standards for such a system.

## 2 Threat Taxonomy: The Poison Data Landscape

We classify data poisoning attacks into four structural types, ordered by detection difficulty:

### 2.1 Type I: Publisher-Level Poisoning

Fabricated papers published through predatory journals, hijacked conference proceedings, cloned journal titles, and fake ISSN registrations. These are the most common and easiest to detect—the publishing channel itself provides strong signals.

### 2.2 Type II: Author-Level Poisoning

Papers from fabricated authors, identity-theft cases (real institution, fake role), impossible publication timelines (35 papers/year), and closed co-author circles where a fixed group of authors only cite and co-publish with each other.

### 2.3 Type III: Cluster-Level Poisoning

The most sophisticated and dangerous form. Multiple papers form a self-referential citation network across different publishers and journals. Individually, each paper may appear legitimate. Collectively, they form a closed validation loop with no external verification. This is the “logically perfect, empirically empty” attack pattern.

### 2.4 Type IV: Lone Perfect Poison

A single paper from a genuine author in a legitimate journal, with internally consistent methodology, but privately fabricated experimental data. No cluster exists to detect. This type bypasses publisher, author, and cluster checks—requiring physical reproducibility verification as the final defense.

### 2.5 Type V: Logical Fallacy Embedding

Poisoned reasoning chains that embed formal fallacies (affirming the consequent, circular reasoning), informal fallacies (appeal to authority, false dichotomy), and quantitative fallacies (Simpson’s paradox, survivorship bias) within otherwise well-structured arguments.

Table 1: Detection coverage by poison type and defense layer

| Poison Type         | Tier I (Publisher) | Tier II (Author) | Tier III (Cluster) |
|---------------------|--------------------|------------------|--------------------|
| Predatory journal   | ✓                  | -                | -                  |
| Fabricated authors  | -                  | ✓                | -                  |
| Citation cartels    | -                  | -                | ✓                  |
| Lone perfect poison | -                  | -                | -                  |

## 3 The Four-Tier Detection Chain

### 3.1 Design Principles

The detection chain follows four architectural principles:

- **Open reproducibility:** Any party running the same standards should reach the same conclusion
- **Full traceability:** Every score can be traced back to specific evidence
- **Multi-dimensional cross-validation:** Failure of a single dimension does not cause systemic misjudgment
- **No single point of failure:** The scoring system does not depend on the absolute correctness of any single component

### 3.2 Tier I: Publisher Credibility Scoring

**Weight: 20% of composite score.**

Five sub-dimensions, each scored 0–20:

1. **Index inclusion:** SCI/SSCI/AHCI indexing status. Predatory journals:  $-10$ ; fake ISSN:  $-25$ .
2. **Publisher history:** Years of operation.  $< 3$  years:  $-10$ ;  $< 1$  year:  $-15$ .
3. **Peer review process:** Transparent review procedures. "Rapid publication" promises:  $-20$ .
4. **Editorial board:** Verifiable academic backgrounds. Fake board members:  $-20$ .
5. **Database coverage:** Major academic database inclusion. Google Scholar only:  $-10$ .

**Pass threshold:**  $\geq 60/100$ . Below 60 triggers immediate flagging regardless of other tier scores.

### 3.3 Tier II: Author Profiling

**Weight: 25% of composite score.**

Five sub-dimensions:

1. **Identity verifiability:** ORCID, institutional directory. Unverifiable:  $-10$ .
2. **Domain consistency:** Publication scope concentration.  $\geq 3$  unrelated domains:  $-15$  (single red flag).
3. **Publication timeline:** Temporal plausibility. Anomalous productivity ( $> 1$  paper/week sustained):  $-10$ . Future citations:  $-20$ .
4. **Co-author network:** Verifiable collaborators. Circular co-author loops:  $-20$ .
5. **Academic trajectory:** Career progression coherence. Sudden domain jumps without transition:  $-15$ .

**Pass threshold:**  $\geq 60/100$ .

### 3.4 Tier III: Cluster Detection

**Weight: 30% of composite score.**

This is the most critical detection layer. While Tiers I–II examine individual papers, Tier III examines *relationships* between papers:

1. **Publisher clustering:**  $\geq 3$  related papers from the same publisher:  $-10$ .
2. **Author circle clustering:** High co-authorship/citation overlap within a small group:  $-15$ .
3. **Temporal clustering:** Concentrated publication windows:  $-15$ .
4. **Citation loops:** Mutual citation forming closed loops:  $-20$ ; multi-level loops:  $-25$ .
5. **Content fingerprinting:** Highly consistent writing patterns across supposedly different authors:  $-15$ .

**Key insight:** A single paper with mediocre Tier I–II scores may be merely low-quality. Multiple papers forming a detection cluster is strong evidence of deliberate fabrication—regardless of individual tier scores.

### 3.5 Tier IV: Cross-Cluster Gang Identification

**Weight: 25% of composite score.**

Tier IV operates across previously detected clusters to identify coordinated groups:

1. **Cross-domain author mapping:** Overlapping authors across domain-separated clusters:  $-20$ .
2. **Content synergy:** Papers that mutually "prove" each other's claims:  $-25$ ; synergy chains  $\geq 3$  papers:  $-30$ .
3. **Temporal orchestration:** Staggered publication timing suggesting scripted sequencing:  $-15$ .
4. **Funding source convergence:** Multiple papers acknowledging the same non-existent funding body:  $-10$ .
5. **Infrastructure sharing:** Same GitHub org, same submission server, identical acknowledgment templates:  $-10$ .

### 3.6 Composite Verdict Matrix

Table 2: Composite verdict matrix across four tiers

| Tier I    | Tier II   | Tier III  | Tier IV       | Verdict     |
|-----------|-----------|-----------|---------------|-------------|
| $\geq 70$ | $\geq 70$ | $\geq 60$ | $\geq 60$     | Trusted     |
| 60–69     | 60–69     | 40–59     | 40–59         | Suspect     |
| $< 60$    | $< 60$    | $< 40$    | $< 40$        | Poison      |
| any       | any       | any       | $\geq 2$ hits | Gang Poison |

## 4 The Five-Anchor Audit Framework

Single-source verification is structurally unreliable. Any single data point, no matter how credible its source, may be wrong. The five-anchor framework ensures that every piece of information is cross-validated across five independent dimensions:

1. **Source Anchor** ( ): Where does this information come from? What is the chain of custody? Analogous to canonicalization in religious traditions—only texts with verified provenance enter the trusted corpus.
2. **Logic Anchor** ( ): Is the reasoning chain internally consistent? Does it survive adversarial examination? Analogous to the scholastic tradition of disputation, where every proposition must withstand counter-arguments.
3. **Interest Anchor** ( ): Who benefits from this information being accepted as true? Are funding sources, institutional affiliations, and conflicts of interest fully declared and independently verified?
4. **Compliance Anchor** ( ): Does the information comply with established standards? Compliance is not binary—standards evolve from recommendation to requirement based on conditional triggers (scale, external threat, trust crisis, conflict of interest).
5. **Cross Anchor** ( ): Can the same conclusion be reached through multiple independent sources? This anchor is explicitly designed to prevent the single-tribunal problem—history demonstrates that single-authority judgments of “heresy” are systematically unreliable.

The five anchors are not additive—they are multiplicative. One anchor failing to verify is sufficient to flag the information for review, regardless of the other four.

## 5 False Positive Remediation: Auditing the Audit Chain

An immune system that cannot correct its own mistakes is not an immune system—it is an inquisition. We design a four-stage remediation pipeline:

### 5.1 Stage 1: Automatic Hold

When Tier IV (gang identification) triggers without Tier I–III support, the verdict is automatically held pending human review. This prevents the highest-confidence (and highest-risk) judgments from executing without oversight.

### 5.2 Stage 2: Independent Review Tribunal

A panel of  $\geq 3$  reviewers, including at least 1 external expert, conducts a blind re-review. The panel does not see the original scores, preventing anchoring bias.

### 5.3 Stage 3: Graduated Recovery

Verdicts are not binary (poison/clean). Recovery follows a five-level gradient:

- **Confirmed Poison:** Appeal rejected, mark retained

- **Downgraded to Suspect:** Evidence insufficient for poison, but doubts remain
- **Insufficient Evidence:** Original judgment excessive, mark removed
- **Clear False Positive:** Judgment error, mark removed with credit restoration
- **New Evidence Override:** New information exonerates, no fault on original judge

## 5.4 Stage 4: Meta-Audit Logging

Every appeal and its outcome is recorded in an immutable meta-audit log. Annual external audits sample 5% of meta-audit logs. Patterns of systemic misjudgment trigger standard revision, not individual case reversal—preserving finality while enabling evolution.

## 5.5 Reviewer Bias Detection

Individual reviewer bias is monitored through a complexity-adjusted deviation coefficient:

$$D_{\text{corrected}} = \frac{|\text{uphold}_r - \text{uphold}| + |\text{overturn}_r - \text{overturn}|}{\text{complexity}_r / \text{complexity}} \quad (1)$$

where  $\text{complexity}_r$  is the average detection tier level (1–4) of cases assigned to reviewer  $r$ . This prevents reviewers assigned to inherently harder cases (Tier IV gang identifications) from being penalized for lower uphold rates.

# 6 The Three-Line Defense Architecture

No single detection layer can catch all poison. We deploy three complementary lines of defense:

## 6.1 Line 1: Metadata Layer (Four-Tier Chain)

Detects source, author, cluster, and gang anomalies. Covers ~80% of known poison types. Limitation: cannot detect lone perfect poison where metadata is clean.

## 6.2 Line 2: Data Layer (Statistical Detection)

Applies Benford’s Law, digit psychology, variance analysis, duplication detection, temporal consistency checks, and benchmark comparison to detect fabricated statistics and experimental data. Covers statistical forgery but requires access to raw data.

## 6.3 Line 3: Physical Verification Layer (Reproducibility)

The final defense. For high-impact judgments, code replication, statistical re-analysis, experimental reproduction, and data provenance tracing are triggered. This layer provides *terminal verdicts*—empirical results have no interpretative space for further appeal.

Table 3: False negative risk reduction with three defense lines

| Poison Type                         | Line 1 | Line 2 | Line 3 |
|-------------------------------------|--------|--------|--------|
| Paper mill operations               | ✓      | -      | -      |
| Predatory journal singles           | ✓      | -      | -      |
| Citation cartels                    | ✓      | -      | -      |
| Statistical forgery                 | -      | ✓      | -      |
| <b>Lone perfect poison</b>          | -      | ✓*     | ✓      |
| <b>Real data, false conclusions</b> | -      | -      | ✓      |

\*Requires access to raw data. If data is private, marked "unverifiable."

## 7 Unverifiable Content: Lifecycle Management

When original data is unavailable (e.g., proprietary datasets), content is marked **Unverifiable**—not **Poison**. This is a critical distinction. Unverifiable means "we cannot confirm," not "we have confirmed false." The mark includes:

- A 12-month recovery window (from notification acknowledgment date)
- Full credit restoration upon data submission and independent verification
- Automatic escalation to **Data Untrusted** (permanent) after 12 months of non-compliance

This prevents the Unverifiable mark from becoming a de facto permanent distrust mark (false positive by proxy) while maintaining strong incentives for data transparency.

## 8 Discussion

### 8.1 Experimental Roadmap

This paper presents the framework architecture and calibration methodology. The framework has been implemented against a training corpus of 110 annotated samples spanning publisher-level, author-level, cluster-level, and gang-level poisoning attacks, as well as statistical data forgery and logical fallacies. A detailed experimental evaluation—including detection recall/precision across all four tiers, false positive remediation throughput, and cross-reviewer bias analysis—will be presented in a follow-up paper (v2). Preliminary calibration results indicate the four-tier detection chain alone captures 82% of known poison types, with the three-line defense architecture projected to reduce false negative risk from medium-high to low. We release the calibration standards and detection methodology under open-source license to enable independent benchmarking and community contribution.

### 8.2 Future Directions

- **Constitution-as-Code:** The framework standards themselves should be machine-readable and automatically enforceable, enabling AI systems to self-audit against published standards.
- **Decentralized review networks:** Distributed reviewer pools to reduce single-tribunal risk and increase cross-cultural calibration diversity.

- **Automated physical verification:** AI-driven simulation of experimental procedures to detect physically impossible claims without full lab reproduction.
- **Calibration memory sharing:** Cross-organization sharing of anonymized audit logs to build collective immune memory without exposing sensitive data.

### 8.3 Relationship to Existing Work

Our framework complements, rather than replaces, existing AI safety approaches. RLHF, constitutional AI, and adversarial training operate on model outputs; our framework operates on training data ingestion. Together they form a complete immune architecture: data-level defense (our work) plus behavior-level defense (existing work).

## 9 Conclusion

We have presented a multi-layered immune system architecture for defending large language models against coordinated data poisoning. The framework consists of a Four-Tier Detection Chain for identifying poisoned content at the publisher, author, cluster, and gang levels; a Five-Anchor Audit system ensuring no single verification point can be compromised; and a Three-Line Defense architecture that closes the detection loop from metadata to physical reproducibility.

The system is designed to be auditable, appealable, and self-correcting—properties that distinguish an immune system from a filter. We open-source the detection methodology and calibration standards to enable community validation, iterative hardening, and deployment across diverse AI training pipelines.

The question is no longer *whether* training data can be trusted, but *how* trust can be systematically verified, challenged, and restored. The architecture described here provides one answer to that question.

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